

BASIC IMAGE PROCESSING Using MATLAB

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I read 'Create Superlative Images Using OpenCV' software article in September 2015 issue under EFY DVD Plus section and found it to be very good. The article described how image processing can be done using OpenCV library. So I thought about writing an article on basic image processing operations using MATLAB, which is widely used by students and newbies to this field.

MATLAB is a high-performance language for technical computing with powerful commands and syntax. It is used for many purposes like Maths and computation, data analysis, algorithm development, modelling stimulation and prototyping. Edge detection, noise and image histogram modelling are some important and basic topics in image processing.

Image processing using MATLAB

Edge detection. An image is nothing but mapping of intensity of the light reflecting from a scene captured from a camera, and edges are the discontinuity of the scene intensity function. We can detect these edges using MATLAB commands.

There are many methods for edge detection such as Robert's operator, Prewitt operator, Sobel operator, Canny edge detector and so on. Fig. 1 shows edge detection using these operators on cameraman.tif standard image available in MATLAB.

Noise. Noise in any system is unwanted. In image processing, noise in a digital image arises during image acquisition and also during transmission. Different types of

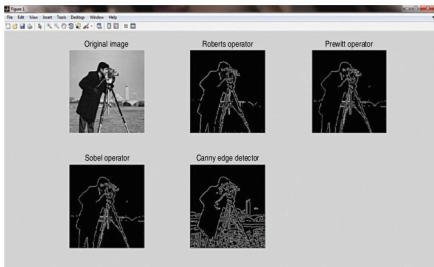


Fig. 1: Edge detection using various operators

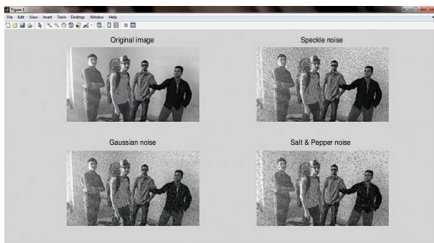


Fig. 2: Special effects in an image using different types of noise

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

noise include speckle, Gaussian, salt-and-pepper and more. The fun part is, we can use these types of noise as special effects in an image using MATLAB.

Fig. 2 shows the results of different types of noise added to an image. In this image, RGB-to-gray